ORGANIC CHEMISTRY ACTIVITY 3

Audrey G. Meteoro

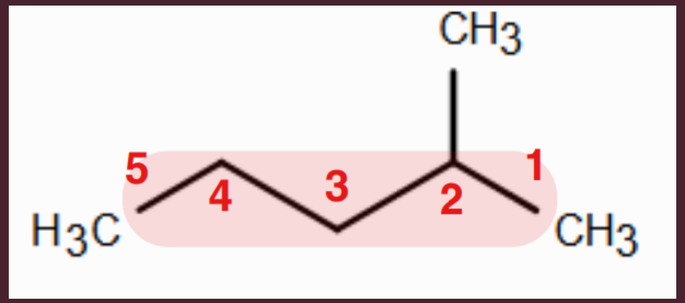
BS ChE 2A

Part A

Part A: CARD A01 - Level: Foundation

Molecular Formula: C_6H_{14}

Numbered Parent:



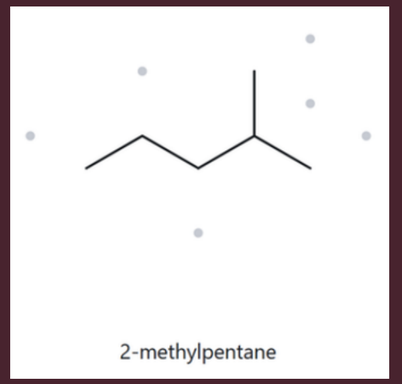
Naming Trail:

Parent: 5 Carbons (Pentane Chain)
Numbering: 5, 4, 3, 2, 1 (from the left)
Substituent: methyl at C-2
Alphabetical: methyl (only present)

My Name:

2 - methylpentane

The Alkane Check:

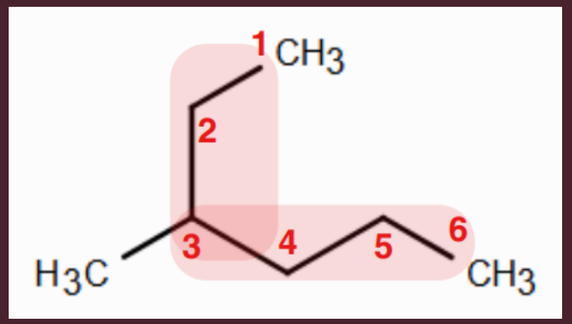


Parent: pentane (5C) | Locants: 2 | Substituent: methyl | Alphabetical: methyl | Formula: C_6H_{14}

Part A: CARD A02 - Level: Foundation

Molecular Formula: C_7H_{16}

Numbered Parent:



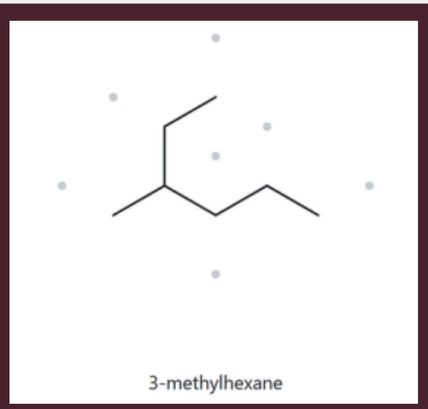
Naming Trail:

Parent: 6 Carbons (Hexane Chain)
Numbering: 1, 2, 3, 4, 5, 6 (from the top-right)
Substituent: methyl at C-3
Alphabetical: methyl (only present)

My Name:

3-methylhexane

The Alkane Check:

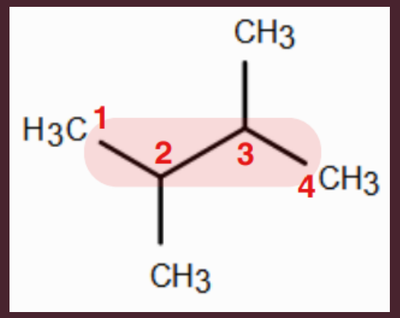


Parent: hexane (6C) | Locants: 3 | Substituent: methyl | Alphabetical: methyl | Formula: C_7H_{16}

Part A: CARD A03 - Level: Foundation

Molecular Formula: C_6H_{14}

Numbered Parent:



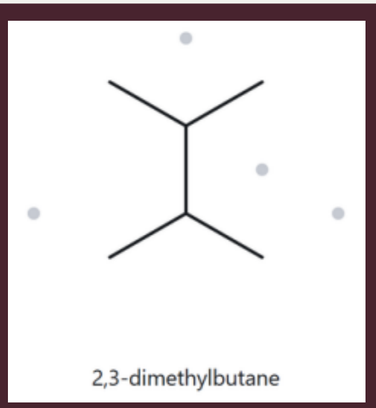
Naming Trail:

Parent: 4 Carbons (Butane Chain)
Numbering: 1, 2, 3, 4 (from the right)
Substituent: methyl at C-2,3
Alphabetical: methyl (only present)

My Name:

2,3-dimethylbutane

The Alkane Check:

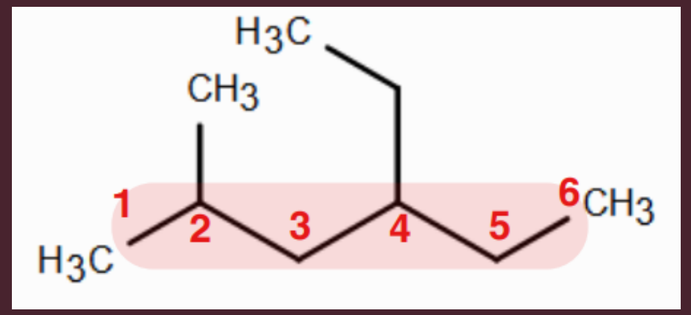


Parent: butane (4C) | Locants: 2,3 | Substituent: methyl | Alphabetical: methyl | Formula: C_6H_{14}

Part A: CARD A04 - Level: Developing

Molecular Formula: C_9H_{20}

Numbered Parent:



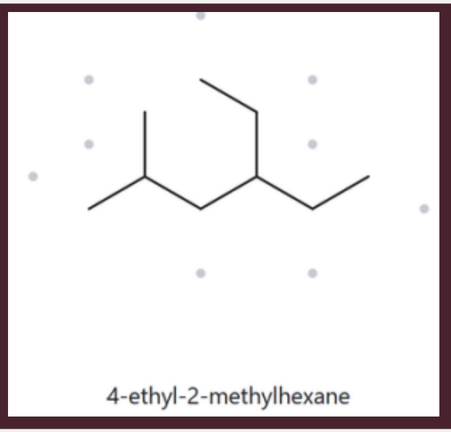
Naming Trail:

Parent: 6 Carbons (Hexane Chain)
Numbering: 1, 2, 3, 4, 5, 6 (from the right)
Substituent: methyl at C-2, ethyl at C-4
Alphabetical: ethyl and methyl

My Name:

4-ethyl-2-methylhexane

The Alkane Check:

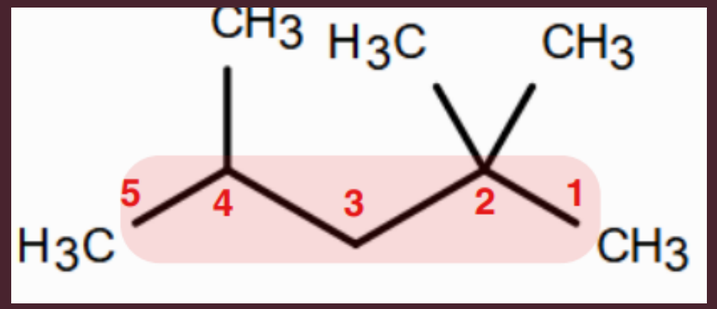


Parent: hexane (6C) | Locants: 2 & 4 | Substituent: methyl & ethyl | Alphabetical: ethyl & methyl | Formula:C_9H_{20}

Part A: CARD A05 - Level: Developing

Molecular Formula: C_8H_{18}

Numbered Parent:



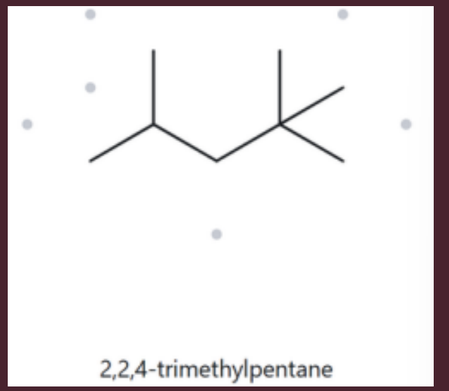
Naming Trail:

Parent: 5 Carbons (Pentane Chain)
Numbering: 5, 4, 3, 2, 1 (from the left)
Substituent: methyl at C-2,2,4
Alphabetical: methyl (only present)

My Name:

2,2,4-trimethylpentane

The Alkane Check:

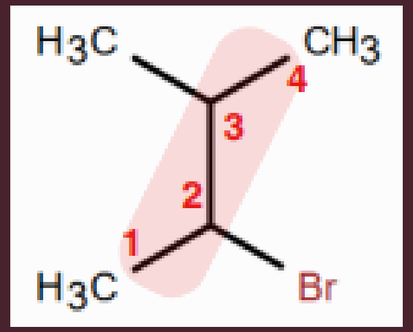


Parent: pentane (5C) | Locants: 2,2,4 | Substituent: methyl | Alphabetical: methyl | Formula: C_8H_{18}

Part A: CARD A06 - Level: Developing

Molecular Formula: $C_5H_{11}Br$

Numbered Parent:



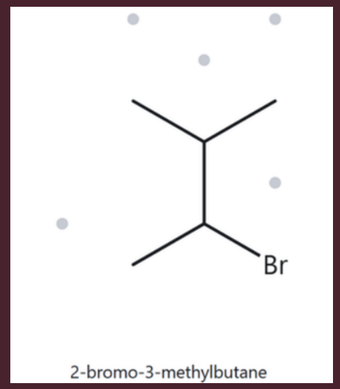
Naming Trail:

Parent: 4 Carbons (Butane Chain)
Numbering: 1, 2, 3, 4 (from the bottom-right)
Substituent: bromo- at C-2, methyl at C-3
Alphabetical: bromo- and methyl

My Name:

2-bromo-3-methylbutane

The Alkane Check:

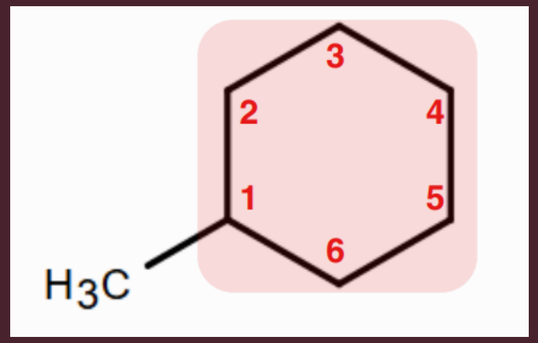


Parent: butane (4C) | Locants: 2 & 3 | Substituent: methyl & bromo- | Alphabetical: bromo- & methyl | Formula: $C_5H_{11}Br$

Part A: CARD A07 - Level: Developing

Molecular Formula: C_7H_{14}

Numbered Parent:



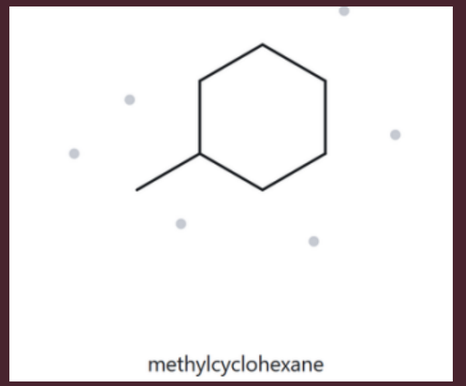
Naming Trail:

Parent: 6 Carbons (Cyclohexane Chain)
Numbering: 1, 2, 3, 4, 5, 6 (from the bottom-right, clockwise)
Substituent: methyl at C-1
Alphabetical: methyl (only present)

My Name:

methylcyclohexane

The Alkane Check:

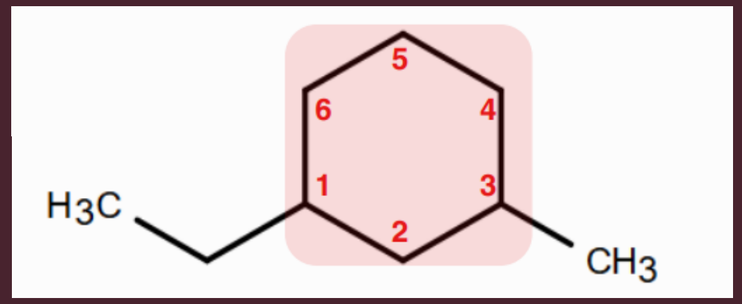


Parent: cyclohexane (6C) | Locants: 1 | Substituent: methyl | Alphabetical: methyl | Formula: C_7H_{14}

Part A: CARD A08 - Level: Challenge

Molecular Formula: C_9H_{18}

Numbered Parent:



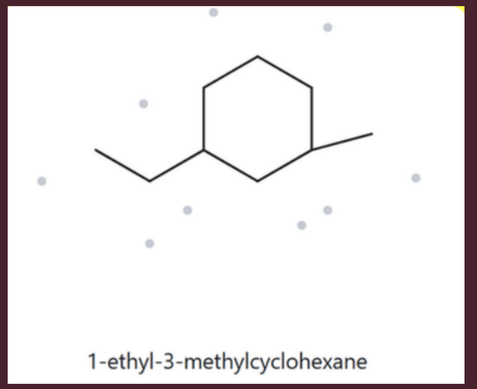
Naming Trail:

Parent: 6 Carbons (Cyclohexane Chain)
Numbering: 1, 2, 3, 4, 5, 6 (from the bottom-right, counter-clockwise)
Substituent: methyl at C-3, ethyl at C-1
Alphabetical: ethyl and methyl

My Name:

1-ethyl-3-methylcyclohexane

The Alkane Check:

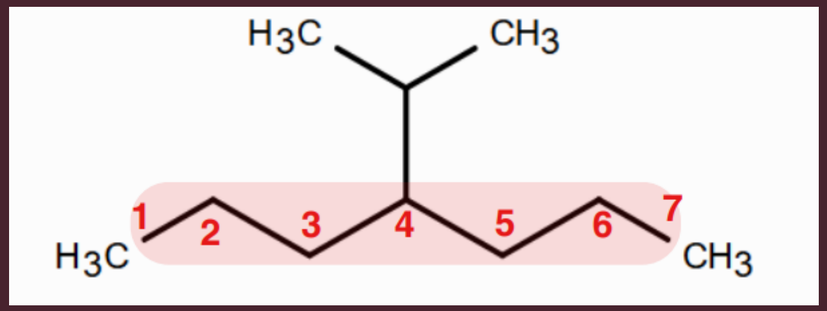


Parent: cyclohexane (6C) | Locants: 1 & 3 | Substituent: methyl & ethyl | Alphabetical: ethyl & methyl | Formula:C_9H_{18}

Part A: CARD A09 - Level: Challenge

Molecular Formula: $C_{10}H_{22}$

Numbered Parent:



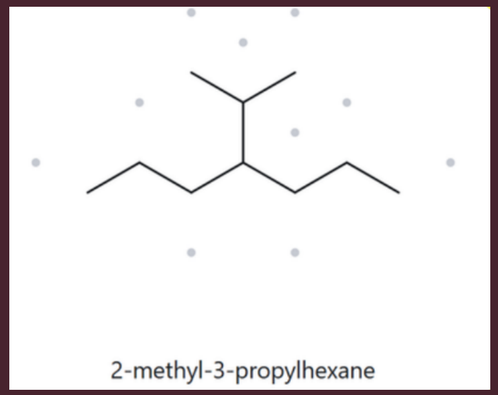
Naming Trail:

Parent: 7 Carbons (Heptane Chain)
Numbering: 1, 2, 3, 4, 5, 6, 7 (from the right)
Substituent: isopropyl at C-4
Alphabetical: isopropyl (only present)

My Name:

4-isopropylheptane

The Alkane Check:

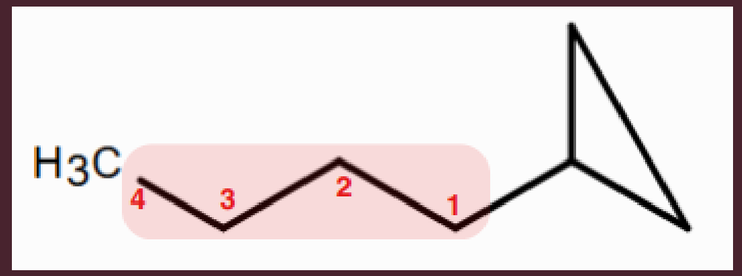


Parent: heptane (7C) | Locants: 4 | Substituent: isopropyl | Alphabetical: isopropyl | Formula: $C_{10}H_{22}$

Part A: CARD A10 - Level: Challenge

Molecular Formula: C_7H_{14}

Numbered Parent:



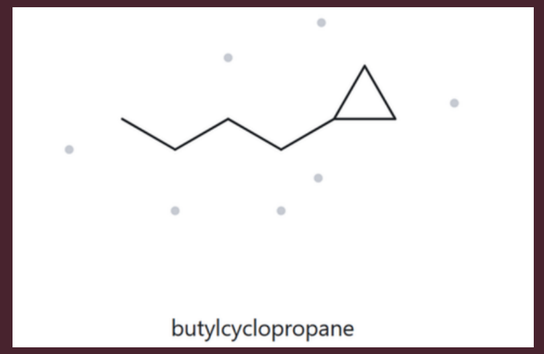
Naming Trail:

Parent: 4 Carbons (Butane Chain)
Numbering: 4, 3, 2, 1 (from the left)
Substituent: cyclopropyl at C-1
Alphabetical: cyclopropyl (only present)

My Name:

butylcyclopropane

The Alkane Check:



Parent: butane (4C) | Locants: 1 | Substituent: cyclopropyl | Alphabetical: cyclopropyl | Formula: C_7H_{14}

ORGANIC CHEMISTRY ACTIVITY 3

Audrey G. Meteoro

BS ChE 2A

Part B

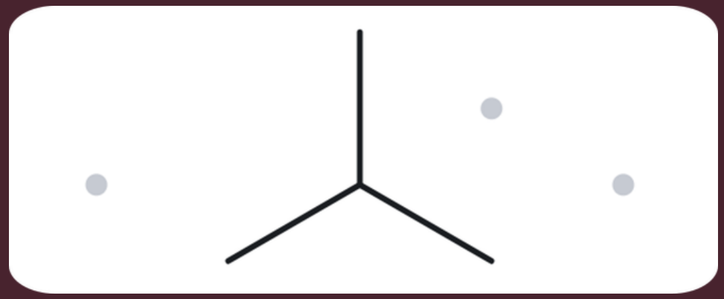
Part B: CARD B01 - Level: Foundation

Given Name: 2-methylpropane

Name Decoded:

Parent: 3 Carbons (Propane Chain)
Substituent: methyl at C-2
Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}

Bond Line Structure:



Formula Check

Counted from the drawing: C_4H_{10}
Open chain: C_nH_{2n+2} = C_4H_{10}

The Back Check:

Returned by The Alkane:
2-methylpropane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

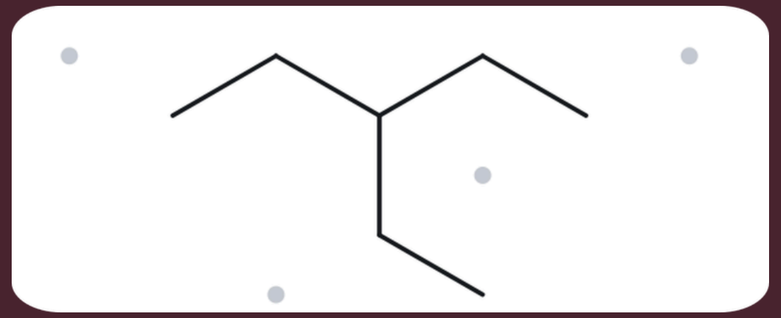
Part B: CARD B02 - Level: Foundation

Given Name: 3-ethylpentane

Name Decoded:

Parent: 5 Carbons (Pentane Chain)
Substituent: ethyl at C-3
Expected formula: 5 + 2 = 7 C; open chain → C_7H_{16}

Bond Line Structure:



Formula Check

Counted from the drawing: C_7H_{16}
Open chain: C_nH_{2n+2} = C_7H_{16}

The Back Check:

Returned by The Alkane:
3-ethylpentane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

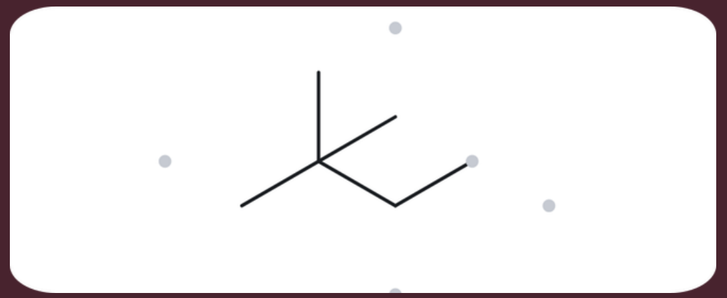
Part B: CARD B03 - Level: Foundation

Given Name: 2,2-dimethylbutane

Name Decoded:

Parent: 4 Carbons (Butane Chain)
Substituent: methyl at C-2,2
Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}

Bond Line Structure:



Formula Check

Counted from the drawing: C_6H_{14}
Open chain: C_nH_{2n+2} = C_6H_{14}

The Back Check:

Returned by The Alkane:
2,2-dimethylbutane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

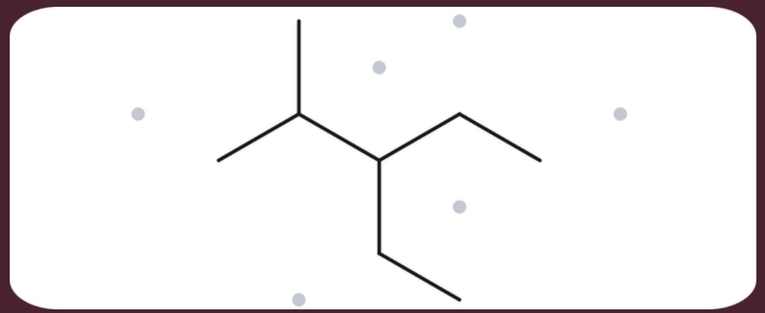
Part B: CARD B04 - Level: Developing

Given Name: 3-ethyl-2-methylpentane

Name Decoded:

Parent: 5 Carbons (Pentane Chain)
Substituent: methyl at C-2, ethyl at C-3
Expected formula: 5 + 2 + 1 = 8 C; open chain → C_8H_{18}

Bond Line Structure:



Formula Check

Counted from the drawing: C_8H_{18}
Open chain: C_nH_{2n+2} = C_8H_{18}

The Back Check:

Returned by The Alkane:
3-ethyl-2-methylpentane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

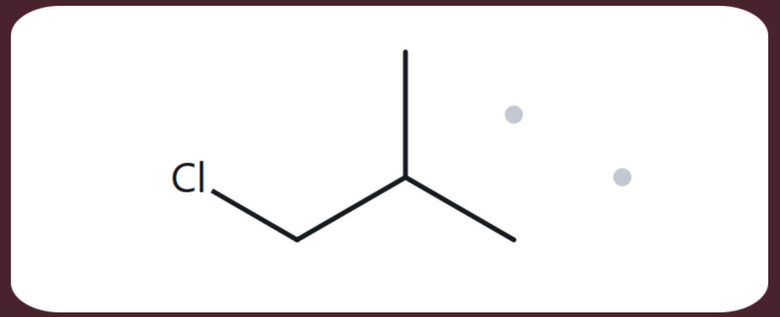
Part B: CARD B05 - Level: Developing

Given Name: 1-chloro-2-methylpropane

Name Decoded:

Parent: 3 Carbons (Propane Chain)
Substituent: methyl at C-3, Chloro- at C-1
Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}; one H replaced by Cl → C_4H_9Cl

Bond Line Structure:



Formula Check

Counted from the drawing: C_4H_9Cl
Open chain: C_nH_{2n+2} = C_4H_{10}
Replace one H with Cl → C_4H_9Cl

The Back Check:

Returned by The Alkane:
1-chloro-2-methylpropane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

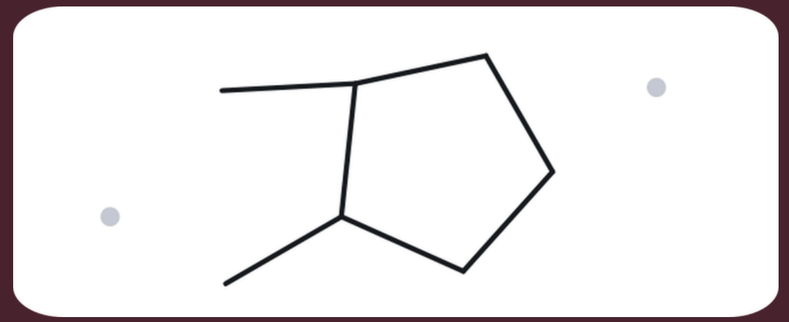
Part B: CARD B06 - Level: Developing

Given Name: 1,2-dimethylcyclopentane

Name Decoded:

Parent: 5 Carbons (Cyclopentane Chain)
Substituent: methyl at C-1,2
Expected formula: 5 + 2 = 7 C; one ring → C_7H_{14}

Bond Line Structure:



Formula Check

Counted from the drawing: C_7H_{14}
One ring: C_nH_{2n} = C_7H_{14}

The Back Check:

Returned by The Alkane:
1,2-dimethylcyclopentane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

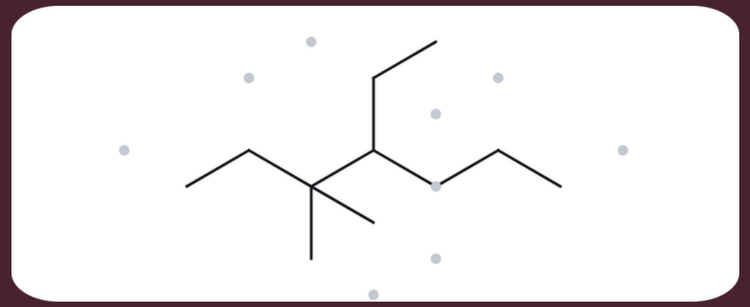
Part B: CARD B07 - Level: Developing

Given Name: 4-ethyl-3,3-dimethylheptane

Name Decoded:

Parent: 7 Carbons (Heptane Chain)
Substituent: methyl at C-3,3, ethyl at C-4
Expected formula: 7 + 2 + 1 + 1 = 11 C; open chain
→ $C_{11}H_{24}$

Bond Line Structure:



Formula Check

Counted from the drawing: $C_{11}H_{24}$
Open chain: C_nH_{2n+2} = $C_{11}H_{24}$

The Back Check:

Returned by The Alkane:
4-ethyl-3,3-dimethylheptane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

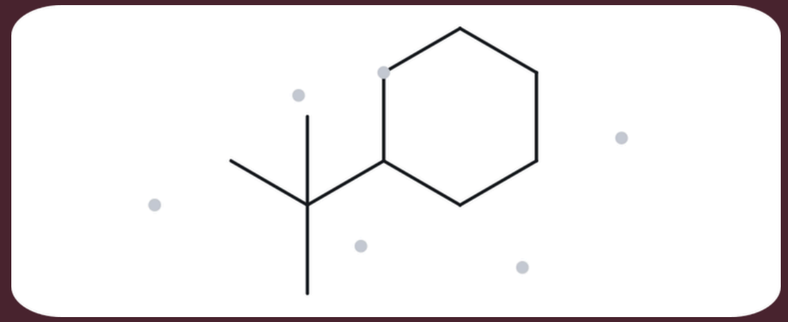
Part B: CARD B08 - Level: Challenge

Given Name: tert-butylcyclohexane

Name Decoded:

Parent: 6 Carbons (Cyclohexane Chain)
Substituent: tert-butyl at C-1
Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$

Bond Line Structure:



Formula Check

Counted from the drawing: $C_{10}H_{20}$
One ring: C_nH_{2n} = $C_{10}H_{20}$

The Back Check:

Returned by The Alkane:
(1,1-dimethylethyl)cyclohexane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

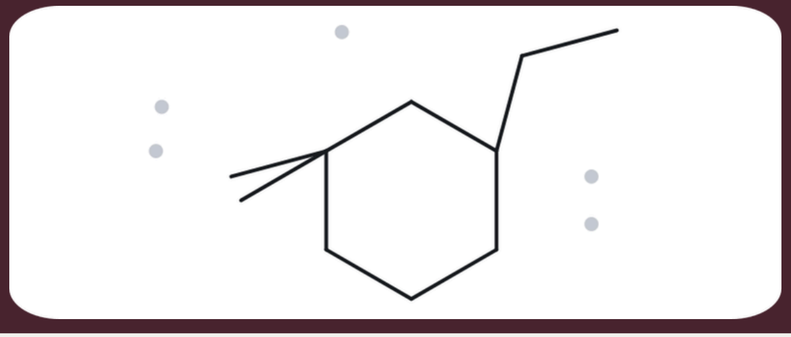
Part B: CARD B09 - Level: Challenge

Given Name: 3-ethyl-1,1- dimethylcyclohexane

Name Decoded:

Parent: 6 Carbons (Cyclohexane Chain)
Substituent: methyl at C-1,1, ethyl at C-3
Expected formula: 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$

Bond Line Structure:



Formula Check

Counted from the drawing: $C_{10}H_{20}$
One ring: C_nH_{2n} = $C_{10}H_{20}$

The Back Check:

Returned by The Alkane:
3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED

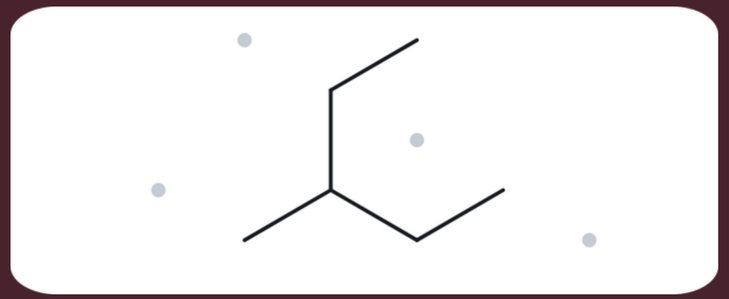
Part B: CARD B10 - Level: Challenge

Given Name: 2-ethylbutane

Name Decoded:

Parent: 5 Carbons (Pentane Chain)
Substituent: methyl at C-3
Expected formula: 4 + 2 = 6, C_6H_{14}

Bond Line Structure:



Formula Check

Counted from the drawing: C_6H_{14}
Open chain: C_nH_{2n+2} = C_6H_{14}

The Back Check:

Returned by The Alkane:
3-methylpentane

Not the same as the given name

PARENT LENGTH MATCHES| SUBSTITUENT POSITIONS MATCH| FORMULA AGREES| BACK-CHECK RECORDED